

$$\begin{aligned} P &= \begin{bmatrix} 5 & 2 \\ 2 & 1 \end{bmatrix} \\ S &= \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \end{aligned} \quad (8.100)$$

P is symmetric positive definite, S is upper triangular, and $P = SS^T$. It can be shown that if S is upper triangular, then $\dot{S}$ and S^{-1} are also upper triangular (see Problem 8.4). Also, the product of upper triangular matrices is another upper triangular matrix (see Problem 8.5). Therefore, the product $S^{-1}\dot{S}$ is upper triangular. Similarly, since $\dot{S}^T$ and S^{-T} are lower triangular, the product $\dot{S}^T S^{-T}$ is lower triangular. That is,

$$\begin{aligned} S^{-1}\dot{S} &= M_U \\ \dot{S}^T S^{-T} &= M_L \end{aligned} \quad (8.101)$$

where M_U and M_L denote upper triangular and lower triangular matrices. From this we can obtain

$$\dot{S} = SM_U \quad (8.102)$$

Now we can use Equations (8.99) and (8.101) to find

$$\begin{aligned} S^{-1}\dot{P}S^{-T} &= S^{-1}\dot{S} + \dot{S}^T S^{-T} \\ &= M_U + M_L \end{aligned} \quad (8.103)$$

So we see that M_U is the upper triangular portion of $S^{-1}\dot{P}S^{-T}$. This gives us the square root algorithm as follows.

The continuous-time square root Kalman filter

1. The initialization step consists of computing the upper triangular $S(0)$ such that $S(0)S^T(0) = P(0)$.
2. At each time step compute $\dot{P}$ from the differential Riccati equation, and then compute M_U as the upper triangular portion of $S^{-1}\dot{P}S^{-T}$.
3. Use $\dot{S} = SM_U$ to integrate S to the next time step.
4. Use the equation $K = PC^T R^{-1} = SS^T C^T R^{-1}$ to compute the Kalman gain.

This is more computationally expensive than a straightforward integration of the differential Riccati equation, but it is also more numerically stable. The numerical benefits of square root filtering are discussed in more detail in Section 6.3.

8.4 GENERALIZATIONS OF THE CONTINUOUS-TIME FILTER

In this section, we will discuss some generalizations of the continuous-time Kalman filter, just as we did in Chapter 7 for the discrete-time Kalman filter. The continuous-time filter was derived under the assumptions that the process and measurement noise was uncorrelated, and that the process and measurement noise was white. We will consider the case in which the process and measurement noise are correlated in Section 8.4.1, and the case in which the measurement noise is colored in Section 8.4.2.

8.4.1 Correlated process and measurement noise

Consider the continuous-time system

$$\begin{aligned}
 \dot{x} &= Ax + w \\
 w &\sim (0, Q) \\
 y &= Cx + v \\
 v &\sim (0, R) \\
 E[w(t)v^T(\tau)] &= M\delta(t - \tau)
 \end{aligned} \tag{8.104}$$

Since $y - Cx - v = 0$ we can write the system dynamics as

$$\begin{aligned}
 \dot{x} &= Ax + w + MR^{-1}(y - Cx - v) \\
 &= (A - MR^{-1}C)x + MR^{-1}y + (w - MR^{-1}v) \\
 &= \tilde{A}x + \tilde{u} + \tilde{w}
 \end{aligned} \tag{8.105}$$

where $\tilde{A}$, $\tilde{u}$, and $\tilde{w}$ are defined by the above equation. Note that $\tilde{u}$ is a known input to the $\dot{x}$ equation, and $\tilde{w}$ is a new process noise term. The cross covariance between the new process noise $\tilde{w}$ and the measurement noise v can be found as

$$\begin{aligned}
 E(\tilde{w}v^T) &= E[(w - MR^{-1}v)v^T] \\
 &= E(wv^T) - MR^{-1}E(vv^T) \\
 &= M - M \\
 &= 0
 \end{aligned} \tag{8.106}$$

So $\tilde{w}$ and v are uncorrelated. The covariance of the new process noise $\tilde{w}$ can be found as

$$\begin{aligned}
 \tilde{Q} &= E(\tilde{w}\tilde{w}^T) \\
 &= E[(w - MR^{-1}v)(w - MR^{-1}v)^T] \\
 &= Q - MR^{-1}M^T - MR^{-1}M^T + MR^{-1}M^T \\
 &= Q - MR^{-1}M^T
 \end{aligned} \tag{8.107}$$

The differential Riccati equation for Kalman filter for the system given in Equation (8.105) is given by

$$\begin{aligned}
 \dot{P} &= \tilde{A}P + P\tilde{A}^T - PC^TR^{-1}CP + \tilde{Q} \\
 &= (A - MR^{-1}C)P + P(A - MR^{-1}C)^T - PC^TR^{-1}CP + \\
 &\quad Q - MR^{-1}M^T
 \end{aligned} \tag{8.108}$$

If we define $\tilde{K}$ as

$$\begin{aligned}
 \tilde{K} &= K + MR^{-1} \\
 &= PC^TR^{-1} + MR^{-1} \\
 &= (PC^T + M)R^{-1}
 \end{aligned} \tag{8.109}$$

then the differential Riccati equation becomes

$$\dot{P} = AP + PA^T + Q - \tilde{K}R\tilde{K}^T \tag{8.110}$$

The differential equation for the state estimate can be written as

$$\begin{aligned}
 \dot{\hat{x}} &= \tilde{A}\hat{x} + \tilde{u} + K(y - C\hat{x}) \\
 &= (A - MR^{-1}C)\hat{x} + MR^{-1}y + K(y - C\hat{x}) \\
 &= A\hat{x} - MR^{-1}C\hat{x} + MR^{-1}y + (\tilde{K} - MR^{-1})(y - C\hat{x}) \\
 &= A\hat{x} + \tilde{K}(y - C\hat{x})
 \end{aligned} \tag{8.111}$$

We see that the introduction of correlation between the process and measurement noise has the effect of simply modifying the Kalman gain. The state-update equation and the differential Riccati equation retain the same form as for the standard Kalman filter. The Kalman filter for correlated process and measurement noise can be summarized as follows.

The continuous-time Kalman filter with correlated noise

1. The system dynamics and measurement equation are given as

$$\begin{aligned}
 \dot{x} &= Ax + w \\
 w &\sim (0, Q) \\
 y &= Cx + v \\
 v &\sim (0, R) \\
 E[w(t)v^T(\tau)] &= M\delta(t - \tau)
 \end{aligned} \tag{8.112}$$

2. The continuous-time Kalman filter is given as

$$\begin{aligned}
 \dot{P} &= AP + PA^T + Q - K RK^T \\
 K &= (PC^T + M)R^{-1} \\
 \dot{\hat{x}} &= A\hat{x} + K(y - C\hat{x})
 \end{aligned} \tag{8.113}$$

Note that (as expected) this filter reduces to the standard continuous-time filter of Equation (8.37) if the process and measurement noise are uncorrelated (i.e., $M = 0$). This filter can therefore be considered as a general formulation of the continuous-time Kalman filter, with the situation $M = 0$ as a special case.

8.4.2 Colored measurement noise

In this section we will derive the Kalman filter when the measurement noise is not white. Suppose we have the system

$$\begin{aligned}
 \dot{x} &= Ax + w \\
 w &\sim (0, Q) \\
 y &= Cx + v \\
 \dot{v} &= Nv + \phi \\
 \phi &\sim (0, \Phi)
 \end{aligned} \tag{8.114}$$

We will assume that w and ϕ are uncorrelated white noise processes. We could augment v onto the state vector (as suggested in Section 7.2.2 for discrete-time systems), but then the covariance of the measurement noise of the augmented system

would be singular, which could potentially cause numerical problems in the Kalman filter implementation. Instead, we will define a new signal as

$$\begin{aligned}
 \tilde{y} &= \dot{y} - Ny \\
 &= \dot{C}x + C\dot{x} + \dot{v} - N(Cx + v) \\
 &= \dot{C}x + C(Ax + w) + (Nv + \phi) - N(Cx + v) \\
 &= (\dot{C} + CA - NC)x + (Cw + \phi) \\
 &= \tilde{C}x + \tilde{v}
 \end{aligned} \tag{8.115}$$

where $\tilde{C}$ and $\tilde{v}$ are defined by the above equation. Note that $\tilde{v}$ is a white noise process (since w and ϕ are uncorrelated and white). So we have defined a new measurement equation that has white noise, but this is at the expense of creating a correlation between the process noise w and the new measurement noise $\tilde{v}$. The correlation can be obtained as

$$\begin{aligned}
 E[w(t)\tilde{v}^T(\tau)] &= E[w(t)(Cw(\tau) + \phi(\tau))^T] \\
 &= QC^T\delta(t - \tau) + 0 \\
 &= M\delta(t - \tau)
 \end{aligned} \tag{8.116}$$

where the cross correlation matrix M is defined by the above equation. The covariance of the new measurement noise $\tilde{v}$ can be obtained as

$$\begin{aligned}
 E(\tilde{v}\tilde{v}^T) &= E[(Cw + \phi)(Cw + \phi)^T] \\
 \tilde{R} &= CQC^T + \Phi
 \end{aligned} \tag{8.117}$$

So we have defined a new measurement equation with white noise. We have the correlation between the process noise and the new measurement noise in Equation (8.116), and the covariance of the new measurement noise in Equation (8.117). Now we can use the results from Section 8.4.1 which discussed Kalman filtering for systems with correlated process and measurement noise. The Kalman filter can be written from Equation (8.113) as

$$\begin{aligned}
 \dot{P} &= AP + PA^T + Q - K\tilde{R}K^T \\
 K &= (PC^T + M)\tilde{R}^{-1} \\
 \dot{\hat{x}} &= A\hat{x} + K(\tilde{y} - \tilde{C}\hat{x}) \\
 &= A\hat{x} + K(\dot{y} - Ny - \tilde{C}\hat{x})
 \end{aligned} \tag{8.118}$$

However, the new measurement that we defined in Equation (8.115) could cause some problems. The original measurement y is already a noisy measurement, so the new measurement (which contains $\dot{y}$) will be even more noisy. How can we avoid the use of $\dot{y}$ in the filter? We can attack this problem by looking at the derivative of the product Ky as follows:

$$\begin{aligned}
 \frac{d(Ky)}{dt} &= \dot{K}y + K\dot{y} \\
 K\dot{y} &= \frac{d(Ky)}{dt} - \dot{K}y
 \end{aligned} \tag{8.119}$$

The dynamic equation for the state estimate in Equation (8.118) can then be written as follows:

$$\begin{aligned}\dot{\hat{x}} &= A\hat{x} + \frac{d(Ky)}{dt} - \dot{K}y - K(Ny + \tilde{C}\hat{x}) \\ \dot{\hat{x}} - \frac{d(Ky)}{dt} &= (A - K\tilde{C})\hat{x} - (\dot{K} + KN)y\end{aligned}\quad (8.120)$$

Now define a new signal z as

$$z = \hat{x} - Ky \quad (8.121)$$

Differentiating z results in the right side of Equation (8.120):

$$\dot{z} = (A - K\tilde{C})\hat{x} - (\dot{K} + KN)y \quad (8.122)$$

Here we have an equation for $\dot{z}$ that we can integrate to solve for z . We can then use our solution for z in Equation (8.121) to solve for $\hat{x}$. So the only signal we have to differentiate in the Kalman filter algorithm is the Kalman gain K , because we need $\dot{K}$ in the computation of $\dot{z}$ above. However, this differentiation should be much easier than differentiating y , because we expect the Kalman gain K to be much smoother than the noisy measurement y . The Kalman filter for the case of colored measurement noise can be summarized as follows.

The continuous-time Kalman filter with colored measurement noise

1. The system and measurement equations are given as

$$\begin{aligned}\dot{x} &= Ax + w \\ w &\sim (0, Q) \\ y &= Cx + v \\ \dot{v} &= Nv + \phi \\ \phi &\sim (0, \Phi)\end{aligned}\quad (8.123)$$

where w and ϕ and uncorrelated white noise processes.

2. Make the following matrix definitions:

$$\begin{aligned}\tilde{C} &= \dot{C} + CA - NC \\ \tilde{R} &= CQC^T + \Phi \\ M &= QC^T\end{aligned}\quad (8.124)$$

3. Initialize the Kalman filter as

$$\begin{aligned}K(0) &= [P(0)C^T + M]\tilde{R}^{-1} \\ z(0) &= \hat{x}(0) - K(0)y(0)\end{aligned}\quad (8.125)$$

4. Integrate P , K , and z using the following equations:

$$\begin{aligned}\dot{P} &= AP + PA^T + Q - K\tilde{R}K^T \\ \dot{K} &= \frac{d}{dt}[(PC^T + M)\tilde{R}^{-1}] \\ \dot{z} &= (A - K\tilde{C})\hat{x} - (\dot{K} + KN)y\end{aligned}\quad (8.126)$$